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Genome-wide identification and expression analysis of Dof gene family members in mulberry trees (*Morus notabilis* L.) under drought stress

Jie Dai¹, Yilei Wu¹, Jiang Liu¹, Ling Wei¹, Yichun Zeng¹, Sanmei Liu¹, Jinchuan Zhang¹, Gang Liu^{1*} and Gaiqun Huang^{1*}

Abstract

Background The DNA-binding with one finger (*Dof*) gene is a classical plant transcription factor with specific conserved structural domains, which plays an important role in the regulation of plant metabolism, root development, and stress defense mechanisms. In recent years, drought has severely jeopardized the growth and development of crops and is one of the main environmental factors limiting plant growth and reducing crop yields. Whether *Dof* family genes play a role in mulberry drought resistance is largely unknown.

Results To investigate the characteristics of the *Dof* gene family members in mulberry trees, 24 members of the mulberry *Dof* gene family were identified in the whole mulberry genome, and bioinformatics tools were employed to analyze the protein physicochemical properties, gene structure, phylogenetic evolution, conserved sequences, interaction networks, promoter cis-acting elements, and expression levels of *Dof* genes in mulberry.

Conclusions Most of the mulberry *Dof* proteins were unstable, all were hydrophilic. Subcellular localization prediction showed that all proteins were localized in the nucleus. The prediction of promoter cis-elements showed that the *MnDof* (*Morus notabilis* L. *Dof*) gene family contained 75 promoter cis-elements for growth and development, hormone, light, and stress responses. Amino acid sequence alignment and phylogenetic tree analysis showed that all 24 *Dof* transcription factors contained zinc finger structural proteins. *MnDof* genes were distributed on all chromosomes except chromosomes 1, 2, and 7. A portion of *MnDof* genes responded to drought stress. These results provide information on the *Dof* gene family in mulberry and provide new insights into the function of *MnDof* genes in perennial plants, laying the foundation for further analysis of the function of these genes.

Keywords Mulberry, *Dof* transcription factor, Bioinformatics analysis, Drought stress

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Background

Transcription factors (TFs), also known as trans-acting factors, are a class of proteins with a special structure that allows them to bind to specific DNA sequences in the promoter regions of target genes to regulate their transcription [1]. TFs are involved in plant responses to abiotic stresses, such as drought, high osmotic stress, and extreme temperatures [2]. They also affect important agronomic traits of crops, such as quality, yield, plant structure, flowering time, and organ development. Based on the type of binding structural domains, TFs are divided into approximately 58 gene types [3], which are further divided into subfamilies based on the number of conserved structural and functional domains. These include types, such as the Dof, MYB, MADS, LBD, SAUR, GRF, bHLH, KNOX, WRKY, and NAC TF families [4, 5].

DNA-binding with one finger (Dof) is an important TF in plants [6], that is encoded by multiple genes [7]. Its N-terminus contains 52 amino acid residues that form a highly conserved C2-C2 single-zinc finger domain. Dof TFs can not only specifically bind to DNA, but also participate in protein interactions; their C-termini have a specific transcriptional regulatory domain, and the diversity of sequences in this region leads to diverse *Dof* gene functions [4–7].

As an important TF in plants, *Dof* is widely involved in plant growth and development [8], metabolic regulation [9], stress defense [10], photosensitivity, pigment signaling, and nitrogen use efficiency [11]. Yanagisawa et al. cloned the first *Dof* gene, *ZmDof1*, from maize; this TF was found to regulate light response and nitrogen assimilation processes [12]. Since then, *Dof* has been reported in many plants, such as rice [13], sorghum [14], grapes [15], apple [16], and pears [17].

Dof genes have been reported to play important roles in regulating plant development and stress responses. For example, in tomato, plants overexpressing the *Dof* gene TDDF1 showed higher resistance to drought, salt stress, and late blight caused by *Phytophthora infestans* [18]. The tomato *Dof* gene *SlDof22* affects ascorbic acid accumulation and salt stress responses [19]. In rice, salt stress inhibits the expression of *OsDof15* in the roots, whereas overexpression of *OsDof15* reduces root sensitivity to salt stress by limiting ethylene biosynthesis [20]. Overexpression of *GhDof1* in cotton resulted in higher salt and cold tolerance [21]. Previous studies have shown that DOF proteins are involved in various biological processes, play important roles in plant growth and development, and are critical for plant responses to stress.

The Mulberry (*Morus notabilis* L.) is an economically important tree species. China has more than 5,000 years of history of mulberry sericulture [22]. In recent years, with a growing interest in expanding the uses of mulberry

beyond sericulture, mulberry trees have become sources [23], for mulberry leaf production for mulberry tea, matcha powder, and mulberry fruit processing to become dried fruits, jams, and juices. The whole body of the mulberry tree, including the roots, stems, leaves, and fruits, all have a certain medicinal value [24]. Mulberry trees have strong vitality and environmental resistance, and TFs are closely related to mulberry resistance features [25].

Although *Dof* family genes are widely reported to affect plant stress tolerance, whether they play a role in drought tolerance in mulberries is largely unknown. Drought severely impairs crop growth and development, limits plant growth, and reduces crop yields [26]. Drought stress can stimulate the expression of several *Dof*, and *SICDFs* genes, members of the tomato *Dof* gene family, are all drought-regulated, and they may be upstream activators of drought stress-responsive pathways [27]. Yu et al. found that most *CsDofs* responded to varying degrees of drought stress in tea trees [28]. However, studies on the DOF family of TFs in mulberry have not been reported. In this study, we used bioinformatics methods to identify and analyze members of the *Dof* gene family in the mulberry whole genome and induced by drought stress in mulberry plants. These studies provide a theoretical basis for further research on the function of *Dof* gene family members in mulberry breeding under extreme conditions.

Methods

Identification of mulberry *Dof* gene family members

The full-length genome sequence of *M. notabilis* and its protein sequences were obtained from the NCBI database (https://www.ncbi.nlm.nih.gov/data-hub/genome/GCF_000414095.1/) and the Genome Database published by Southwest University (<https://morus.biodb.org/morusdb/>) [29, 30]. The Hidden Markov Model (HMM) profile of the DOF gene family (PF02701) was downloaded from the Pfam database (<http://xfam.org>) using TBtools software for searching and identifying DOF family members in the mulberry genome. Sequences of the *Dof* gene family of *Arabidopsis thaliana*, rice, grape, and peach in the homologous evolutionary analyses were obtained from the NCBI database and TAIR (<https://www.arabidopsis.org/>) and published articles [9, 15, 31, 32].

Bioinformatics analysis of the *Dof* gene family of *M. notabilis*

Bioinformatics analysis and subcellular localization analysis of the mulberry *Dof* gene family

Physicochemical properties, such as molecular weight, isoelectric point, hydrophobicity, instability, fatty amino acid index, and amino acid composition of all proteins encoded by the mulberry gene family were analyzed

using the online website ExPASy-ProtParamtool (<https://web.expasy.org/protparam/>).

Subcellular localization analysis of the mulberry DOF protein sequences was performed using the online website UNIPROT (<https://www.uniprot.org/>).

Analysis of cis-acting elements within promoter regions

For the promoter analysis, the 2000-bp region upstream of the transcription start site (ATG) of each *MnDof* gene was selected as the putative promoter sequence. Cis-acting elements were identified using PlantCARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) with stringent parameters ($E\text{-value} \leq 1e-5$), and results were visualized using TBtools.

Protein conserved motif analysis

Conserved motifs of amino acid sequences coding for the *Dof* gene family in the whole mulberry genome were predicted and analyzed using the MEME v7.0.26 online tool (<http://meme-suite.org/tools/meme>), with the following parameters: maximum motif number = 10, motif width = 6–50 amino acids, and $E\text{-value} \leq 1e-10$.

Phylogenetic tree construction and protein evolutionary relationship analysis

Amino acid sequences of Dof proteins from *M. notabilis*, *A. thaliana*, *O. sativa*, *V. vinifera*, and *P. persica* were aligned using ClustalW in MEGA 11.0. A neighbor-joining (NJ) phylogenetic tree was constructed with 1,000 bootstrap replicates, and the Poisson model was selected for distance calculation. Protein evolutionary relationships analyzed with Orthofinder. The assembly of *Morus alba* (GWHDOOO000000000), *Morus alba* var. *atropurpurea* (GWHBOTE000000000), *Morus yunnanensis* (GWHBISP000000000), and The T2T assembly of *Morus mongolica* (GCA_045269455.1) and *Morus notabilis* (GWHCBHX000000000) were downloaded according to the accession number. The protein sequence of *Arabidopsis* (*Arabidopsis thaliana*), rice (*Oryza sativa Japonica*), grape (*Vitis vinifera*), and peach (*Prunus persica*) were download from EnsemblPlants (<https://plants.ensembl.org/index.html>).

Protein interactions analysis

Uploaded DOF Protein Sequence Search Analyzed Protein Interaction Patterns of Mulberry Dof Gene Family Members at the STRING (v12.0; <https://cn.string-db.org/>) online website.

Analysis of chromosome localization and gene duplication events

The chromosome localization map was created using MapInspect v1.0 software, and information on the locations in the mulberry *Dof* gene assembly was extracted

by local BLAST comparison. The corresponding information on mulberry (*M. alba* L.) chromosomes was obtained from related literature [32]. Collinearity analysis: Homologous gene pairs were identified using MCS-canX (default parameters) with genome annotation files. The protein sequences of Dofs were aligned by Mafft v. 7.520 with the L-INS-i algorithm (–localpair –maxiterate 1000). Multiple sequence alignment of protein and corresponding mRNA sequences was then converted into codon alignment using PAL2NAL (v.14.1). nonsynonymous (Ka) and Synonymous(Ks) substitution rates were calculated using KaKs_Calculator v. 2.0 with the YN00 model (transition/transversion ratio = 2.0).

Expression analysis of mulberry Dof family genes under drought stress

Material treatment and sampling

Seeds of the Guangxi mulberry cultivar 'Gui You 62' were obtained from the germplasm repository of the Sericultural Research Institute, Sichuan Academy of Agricultural Sciences. Uniform seeds were surface-sterilized with 75% ethanol for 1 min, rinsed three times with distilled water, and germinated in peat:vermiculite (3:1) medium. Seedlings were grown in a controlled-environment chamber (16-h light/8-h dark, 25 °C, 65–75% relative humidity) until reaching 10 cm height, then transplanted into individual pots (15 cm diameter) containing nutrient soil.

After 45 days of growth, 60 uniformly sized seedlings were divided into two groups ($n=30$): Drought stress group: Treated with 20% (w/v) PEG6000 solution for hydroponics; Control group: Irrigated with Hoagland's nutrient solution. Root and leaf samples were collected at 0, 3, 6, 12, and 24 h post-treatment. Three biological replicates (individual plants) were collected per time point, immediately frozen in liquid nitrogen, and stored at -80°C .

qRT-PCR analysis

Total RNA was extracted using an OMEGA Plant RNA Kit (R6827-01), with genomic DNA removed by DNase I treatment. RNA integrity was verified by 1% agarose gel electrophoresis, and concentration was measured using a NanoDrop 2000 (Thermo Fisher). cDNA was synthesized from 1 μg RNA using a Takara PrimeScript RT Kit (RR047A).

qRT-PCR was performed on a Bio-Rad CFX96 system with SYBR® Premix Ex Taq™ II (RR820A). Primers (Supplementary Table 1) were designed using Primer-BLAST (NCBI) with the following criteria: amplicon length = 100–200 bp, $T_m = 60 \pm 2^{\circ}\text{C}$, and GC content = 40–60%. *MnActin1* (GenBank: XM_010112836) was used as the internal control. Reaction conditions: 95 °C for 30 s; 40 cycles of 95 °C for 5 s, 60 °C for 30 s. Melting curve analysis (65–95 °C) confirmed primer

specificity. Relative expression was calculated using the $2^{-\Delta\Delta C_t}$ method.

Complies with international, national, and/or institutional guidelines

Experimental research and field studies on plants (either cultivated or wild) comply with relevant institutional, national, and international guidelines and legislation.

Results

Identification and physicochemical characterization of the *Dof* gene family in mulberry genome

A total of 24 non-redundant *Dof* gene family members were identified in the *M. notabilis* (Table 1), and were sequentially named *MnDof1*-*MnDof24*. The results of physicochemical characterization of the DOF protein family members of *M. notabilis* showed that the smallest DOF protein was *MnDof1*, with 167 amino acids, and the largest one was the *MnDof24* protein with 543 amino acids. The aliphatic amino acid index ranged from 36.42 to 62.05; the molecular weight of the proteins ranged from 18.69 to 59.01 kDa, with a maximum isoelectric point of 9.4 (*MnDof1*) and a minimum isoelectric point of 5.04 (*MnDof3*); the instability coefficients showed that the majority of the proteins were unstable (instability coefficients > 40), and only one protein (*MnDof15* = 39.31) was stable; the hydrophilicity values of all mulberry DOF proteins were negative, indicating that all 24 proteins were hydrophilic. The results of subcellular localization predicted nuclear localization for all MnDOF proteins, supporting their roles as transcription regulators.

Promoter sequence analysis

To investigate the transcriptional regulation and other potential functions of *MnDof* genes, the promoter sequences of 24 mulberry *Dof* genes were explored for cis-acting elements. A total of 75 cis-acting elements were predicted (Fig. 1). These were classified into six categories: conventional promoter elements, phytohormone-related response elements, stress response elements, light response elements, growth and development-related elements and functionally unknown elements. Related data is provided in Supplementary Table 2. All *MnDof* genes contained two conventional promoter elements (a TATA-box and a CAAT-box). The most numerous were plant hormone-related response elements, with a total of 401. All *MnDof* genes contained a total of 162 methyl jasmonate (MeJA) response elements (TGACG-motif, CGTCA-motif), indicating that *MnDof* genes are closely related to jasmonic acid signaling in mulberry; 137 abscisic acid (ABA) response elements (AAGAA-motif, ABRE, etc.), 51 ethylene response elements (ERE), 22 salicylic acid (SA) response elements (TCA, TCA-element), 16 auxin response elements, and 13 gibberellin response elements. There are 177 light response elements, primarily including G-box, I-box, GT1-box, TCT-motif, GATA-motif, ATCT-motif, etc. There are 144 stress-responsive elements, including 64 ARE promoter elements, present in all *MnDof* genes except *MnDof2*, *MnDof21*, and *MnDof23*, which are crucial for anaerobic induction; 55 stress-tolerance and disease-resistance elements (as-1, TC-rich repeats); 22 drought-responsive elements (F-box, DRE core, DRE1,

Table 1 The information about the *MnDof* gene family in *Morus notabilis* L

Gene Name ^a	Sequence ID ^a	GenBank ^a	Amino acid-length ^a	Molecular weight ^a	Isoelectric-point (pI) ^a	Instability-index ^a	Aliphatic-index ^a	Hydropathicity ^a	Subcellular location ^a
<i>MnDof1</i> ^a	L484_005785 ^a	EXB94628.1 ^a	167 ^a	18.69 ^a	9.4 ^a	49.91 ^a	56.59 ^a	-0.666 ^a	Nucleus ^a
<i>MnDof2</i> ^a	L484_008109 ^a	EXB34849.1 ^a	193 ^a	21.05 ^a	9.02 ^a	60.26 ^a	36.42 ^a	-0.987 ^a	Nucleus ^a
<i>MnDof3</i> ^a	L484_001828 ^a	EXB91954.1 ^a	221 ^a	25.05 ^a	5.04 ^a	62.97 ^a	58.19 ^a	-0.781 ^a	Nucleus ^a
<i>MnDof4</i> ^a	L484_014436 ^a	EXB38622.1 ^a	246 ^a	26.17 ^a	6.43 ^a	51.44 ^a	43.25 ^a	-0.813 ^a	Nucleus ^a
<i>MnDof5</i> ^a	L484_023476 ^a	EXB83869.1 ^a	274 ^a	30.06 ^a	9.27 ^a	56.5 ^a	52.37 ^a	-0.769 ^a	Nucleus ^a
<i>MnDof6</i> ^a	L484_016920 ^a	EXC12989.1 ^a	297 ^a	32.7 ^a	6.13 ^a	43.38 ^a	62.05 ^a	-0.553 ^a	Nucleus ^a
<i>MnDof7</i> ^a	L484_014490 ^a	EXC18090.1 ^a	310 ^a	34.06 ^a	9.18 ^a	74.89 ^a	39.06 ^a	-1.032 ^a	Nucleus ^a
<i>MnDof8</i> ^a	L484_012119 ^a	EXB82806.1 ^a	319 ^a	35.66 ^a	7.22 ^a	60.67 ^a	57.18 ^a	-0.792 ^a	Nucleus ^a
<i>MnDof9</i> ^a	L484_003636 ^a	EXC07927.1 ^a	323 ^a	35.59 ^a	8.65 ^a	54.12 ^a	57.34 ^a	-0.76 ^a	Nucleus ^a
<i>MnDof10</i> ^a	L484_014430 ^a	EXB98585.1 ^a	332 ^a	36.62 ^a	7.05 ^a	42.13 ^a	55.54 ^a	-0.784 ^a	Nucleus ^a
<i>MnDof11</i> ^a	L484_016668 ^a	EXB89108.1 ^a	342 ^a	37.26 ^a	7.28 ^a	62.15 ^a	51.43 ^a	-0.72 ^a	Nucleus ^a
<i>MnDof12</i> ^a	L484_014489 ^a	EXC18089.1 ^a	348 ^a	37.05 ^a	8.45 ^a	64.16 ^a	56.06 ^a	-0.728 ^a	Nucleus ^a
<i>MnDof13</i> ^a	L484_015022 ^a	EXB87891.1 ^a	351 ^a	38.18 ^a	6.58 ^a	40.8 ^a	54.96 ^a	-0.688 ^a	Nucleus ^a
<i>MnDof14</i> ^a	L484_025208 ^a	EXC21409.1 ^a	354 ^a	38.13 ^a	9.3 ^a	60 ^a	57.06 ^a	-0.628 ^a	Nucleus ^a
<i>MnDof15</i> ^a	L484_011851 ^a	EXC21409.1 ^a	374 ^a	42.2 ^a	7.92 ^a	39.31 ^a	53.72 ^a	-0.996 ^a	Nucleus ^a
<i>MnDof16</i> ^a	L484_011406 ^a	EXC20162.1 ^a	377 ^a	40.46 ^a	9.08 ^a	58.47 ^a	55.65 ^a	-0.702 ^a	Nucleus ^a
<i>MnDof17</i> ^a	L484_021308 ^a	EXC31006.1 ^a	377 ^a	39.54 ^a	8.93 ^a	50.82 ^a	54.99 ^a	-0.538 ^a	Nucleus ^a
<i>MnDof18</i> ^a	L484_026802 ^a	EXC35495.1 ^a	387 ^a	40.96 ^a	9.1 ^a	45.04 ^a	59.07 ^a	-0.561 ^a	Nucleus ^a
<i>MnDof19</i> ^a	L484_020268 ^a	EXB37212.1 ^a	401 ^a	41.81 ^a	8.91 ^a	49.42 ^a	60.92 ^a	-0.478 ^a	Nucleus ^a
<i>MnDof20</i> ^a	L484_000659 ^a	EXB22882.1 ^a	424 ^a	46.2 ^a	5.96 ^a	53.22 ^a	45.57 ^a	-0.817 ^a	Nucleus ^a
<i>MnDof21</i> ^a	L484_008599 ^a	EXB38571.1 ^a	496 ^a	53.98 ^a	6.68 ^a	59.39 ^a	52.2 ^a	-0.864 ^a	Nucleus ^a
<i>MnDof22</i> ^a	L484_002979 ^a	EXC36163.1 ^a	496 ^a	55.19 ^a	6.39 ^a	56.2 ^a	59.19 ^a	-0.776 ^a	Nucleus ^a
<i>MnDof23</i> ^a	L484_015665 ^a	EXC19990.1 ^a	497 ^a	53.59 ^a	6.24 ^a	48.35 ^a	59.24 ^a	-0.672 ^a	Nucleus ^a
<i>MnDof24</i> ^a	L484_013586 ^a	EXB31954.1 ^a	543 ^a	59.01 ^a	5.24 ^a	57.63 ^a	50.87 ^a	-0.865 ^a	Nucleus ^a



Fig. 1 Cis-elements in the promoter regions of Dof genes in mulberry. Each colored block represents functionally similar promoter cis-acting elements

MBS) 22; temperature response elements (CTAG-motif, CCAAT-box) 3, present only in *MnDof1* and *MnDof14*; and growth and development-related elements 231, present in all genes. This cis-element landscape highlights the *MnDof* family's multifunctional roles in stress adaptation and developmental plasticity.

Conserved amino acid sequence motif analysis

To obtain the conserved amino acid information of the mulberry *Dof* gene family members, the amino acid sequences of 24 members were compared across multiple sequences (Fig. 2), and all of them were found to contain a highly conserved typical Dof gene family sequence, Motif1 (CPRC TKFCY NNY QPR FC C R WT GG R P G G), corresponding to the C2-C2 zinc finger structure essential for DNA binding. Further detailed analysis of the conserved motifs of mulberry DOF proteins by MEME showed that they contain 10 independent conserved motifs, named Motif 1–10 (Fig. 3). Motif 1 is a single C2-C2 zinc finger structure present in all mulberry DOF proteins, confirming its role as a signature domain for Dof TF function. Motif 2- Motif 4 flanked Motif1 in groupA, potentially involved in auxiliary DNA recognition. *MnDof21-MnDof24* exhibited unique C-terminal

motifs (Motif 8- Motif 10), containing repetitive proline/ glutamine residues (e.g., Motif 8: P-Q-P-Q), suggesting roles in protein–protein interactions. These results suggest that DOF proteins containing the same conserved motifs in the same group may have similar functions, and MnDOF24 may be associated with protein folding or protein complex assembly.

Evolutionary analysis of Dof gene family members in mulberry and related species

To analyze the affinities between members of the *Dof* gene family in mulberry and other species, the *Dof* genes of four other plants, including *Arabidopsis thaliana*, rice, grapevine, and peach, were selected to construct an evolutionary tree, with genes clustering into six subgroup (Fig. 4). Group I had the most members and contained seven genes, and Group II and IV had the fewest members and contained three genes. While Group VI exclusively contained non-mulberry orthologs.

The results of protein evolutionary analysis (Table 2) were showed that 18 orthogroups were identified, including OG0000774 with *MnDof14*, *MnDof16*, *MnDof17*, *ATDof2.4*, *ATDof3.6*, *OsDof-16*, *OsDof-19*, *OsDof-29*, *PpeDof4*, *PpeDof5*, *PpeDof6*, *VviDofL14*, *VviDofL15*,

DOF domian

MnDOF20	ALKKPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGTLRNPVVGACRRRNKHSSSQY...	68
MnDOF24	TLKKPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGTLRNPVVGACRRRNKHNSASH...	67
MnDOF2	AHFPECEQLKCPROCSSTNTKFCYFNNYNLSCPRHFCNKQCRRYVTGKGALRNIPVGGGSRKNKAK...	63
MnDOF19	TSSSNSENLRCPROCSSNTTKFCYFNNYNLTCPRHFCNKQCRRYVTGKGALRNIPVGGGSRKNKNSITVVSASA	71
MnDOF21	TLKKPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGTLRNPVVGACRRRNKHNSSSH...	67
MnDOF4	GM.GHPPPLPCPRQCSSTNTKFCYFNNYNLACPRHFCNKQCRRYVTGGLRNPVVGGSRKNSKRSRS...	66
MnDOF8	CNCNQSEPLKCPROCSSTNTKFCYFNNYNKSCPRHFCNKQCRRYVTGKGTLRNPVVGGSRKNSKRNKKNK...	67
MnDOF5	ARPCEQLNCPROCSSTNTKFCYFNNYNLTCPRYFCKTCORRYVTGGLRNPVVGGSRKNNKKSITN...	65
MnDOF13	LRPCCDQALKCPROCSSTNTKFCYFNNYNLSCPRYFCKTCORRYVTGGLRNPVVGGSRKNNKKSITSSK...	67
MnDOF11	HHCPDQALKCPROCSSTNTKFCYFNNYNLSCPRHFCNKQCRRYVTGKGVLRNIPVGGGCRKTKRSKPKASSA	71
MnDOF3	VKPVIEAPPCCPRQASSNTKFCYFNNYNLSCPRYFCKTCORRYVTGKGSLRNIPVAGGCRKKNKRSKGR...	67
MnDOF1	ENKRPPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGALRNIPVVGACRRRNKHNSIKPP...	64
MnDOF10	RP.HPEQALKCPROCSSTNTKFCYFNNYNLTCPRYFCKTCORRYVTGGLRNPVVGGSRKNNKRSSSS...	66
MnDOF9	ARPCKEQALNCPROCSSTNTKFCYFNNYNLTCPRYFCKTCORRYVTGGSALRNIPVGGGSRKNKRSSSS...	67
MnDOF14	KNPCEQALNCPROCSSTNTKFCYFNNYNLSCPRHFCNKQCRRYVTGKGALRNIPVGGGCRKNKRSKSG...	67
MnDOF6	VKPNVEISPNCPRQCSSTNTKFCYFNNYNLTCPRYFCKTCORRYVTGKGSLRNIPVGGGCRKNRRKSKSL...	67
MnDOF12	QLCCQCCALKCPROCSSTNTKFCYFNNYNLSCPRHFCNKQCRRYVTGGLRNPVVGGSRKNNKRSKRP...	67
MnDOF7	QPPPHCPCKCPROCSSTNTKFCYFNNYNLSCPRYFCKTCORRYVTGGLRNPVVGGSRKNNKRSKRAK...	65
MnDOF23	TLRKPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGTLRNPVVGACRRRNKHNSSSH...	68
MnDOF16	KLCPDQALKCPROCSSTNTKFCYFNNYNLSCPRHFCNKQCRRYVTGKGALRNIPVGGGCRKNKRSKSG...	66
MnDOF15	ARPCKDQALNCPROCSSTNTKFCYFNNYNLTCPRYFCKTCORRYVTGGSALRNIPVGGGSRKNKRSSSS...	67
MnDOF17	NI.PNPETALKCPROCSSTNTKFCYFNNYNLTCPRHFCNKQCRRYVTGKGALRNIPVGGGCRKNKRSKGT...	67
MnDOF18	SSRPSEQALKCPROCSSTNTKFCYFNNYNLTCPRHFCNKQCRRYVTGKGALRNIPVGGGCRKNKKSITK...	65
MnDOF22	VFKKPEKYLPCPRQNSMDTKFCYFNNYNVNCPRHFCNKQCRRYVTAGGTLRNPVVGACRRRNKHNSHLASHF...	68

Fig. 2 Amino acid sequence alignment

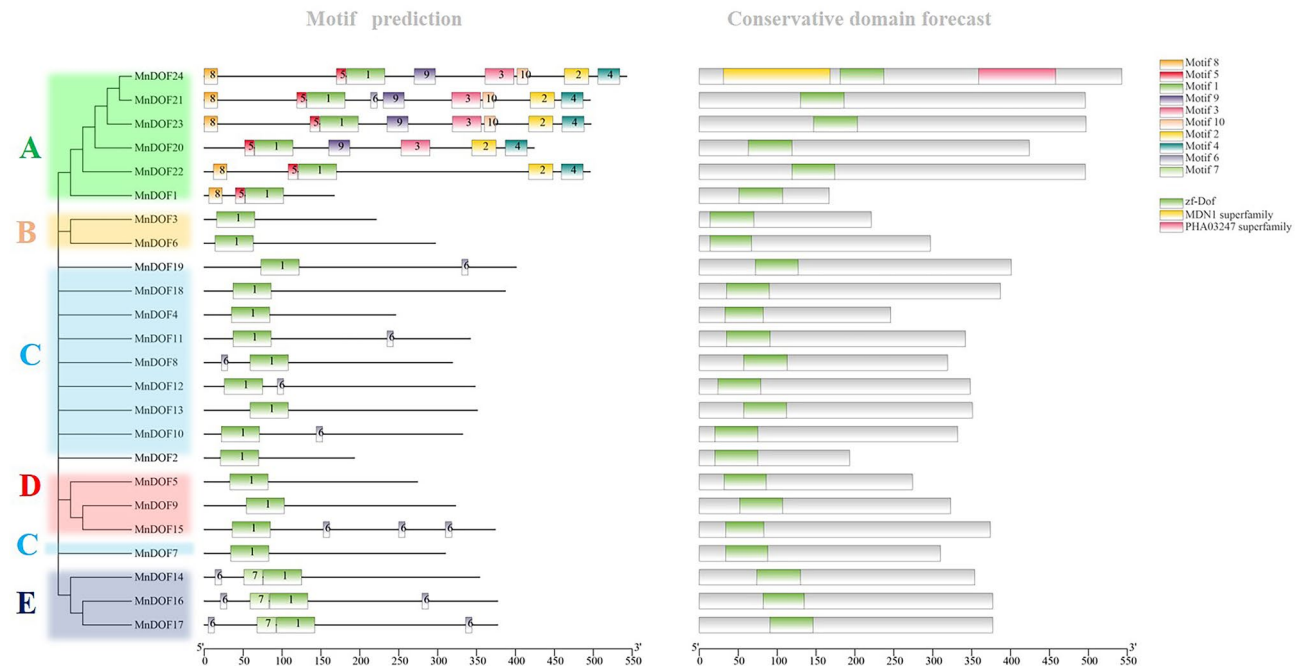


Fig. 3 Phylogenetic (I), conservative motif (II), and conservative domain (III) prediction of mulberry Dof gene family. **A, B, C, D,** and **E** are the five groups of the Dof gene in mulberry

VviDofL16, and also they all belong to the Group I in the phylogenetic tree. The OG0001111 includes *MnDof9*, *MnDof5*, *AtDOF1.8*, *AtDof2.5*, *ATDof3.7*, *ATDof4.6*, *PpeDof15*, *PpeDof13*, *VviDofL11*, *VviDofL13*, *VviDofL18*, and

they are also in the Group III of the tree. OG0001265 has *MnDof21*, *MnDof24*, *MnDof23*, *ATDof3.3*, *ATDof5.2*, *OsDof-11*, *PpeDof22*, *PpeDof21*, *PpeDof20*, and *VviDofL7*, *VviDofL8*, *VviDofL6*, and all of them are in Group V of the

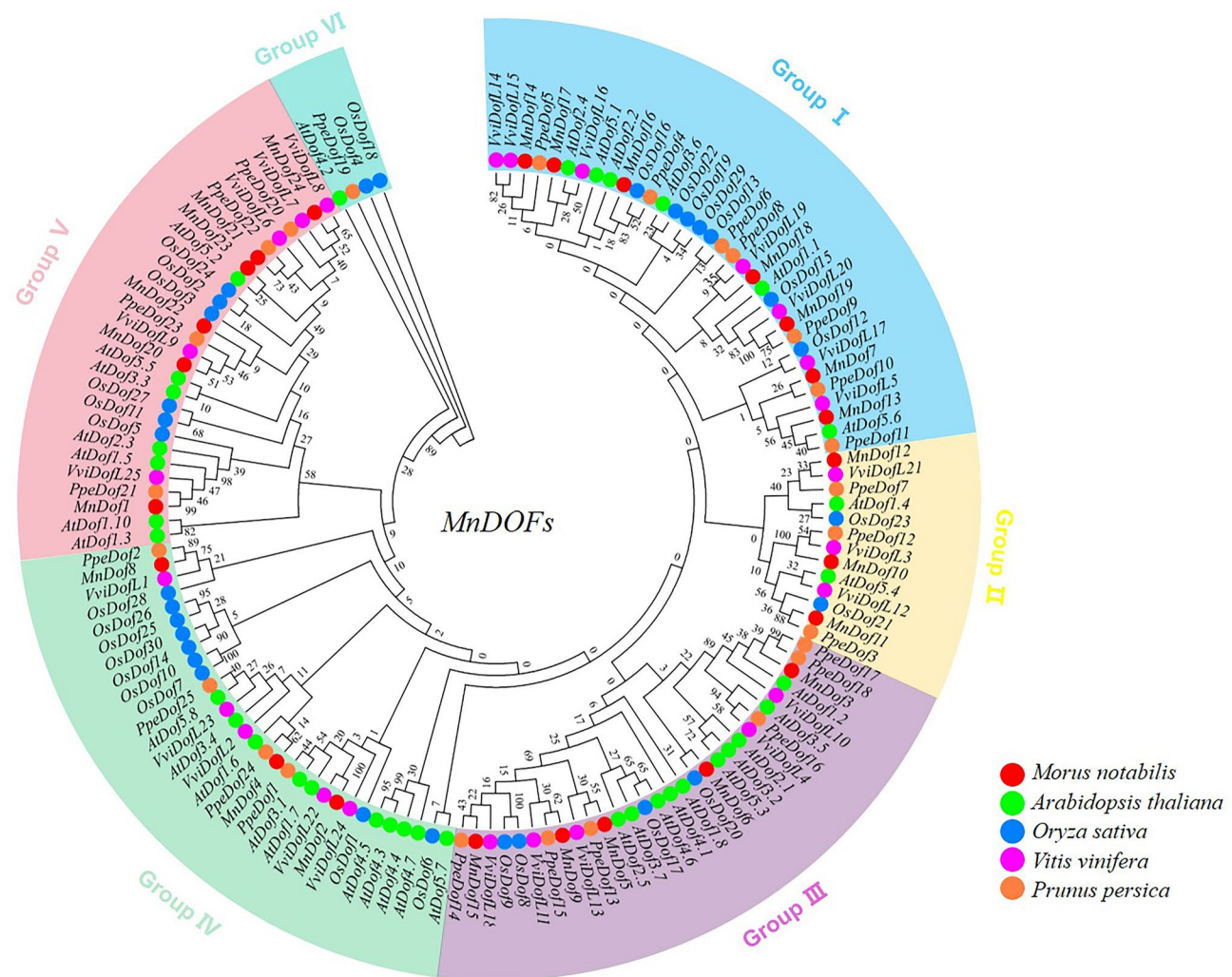


Fig. 4 Phylogenetic tree of mulberry Dof genes. The tree was produced using MEGA11.0 software. The full length of amino acid sequences of 140 DOF proteins (24, *Mn*; 36, *At*; 30, *Os*; 25, *Vv*; 25, *Pp*) were used to generate the evolutionary tree by maximum likelihood method. The DOF proteins of all species were categorized into six groups according to their evolutionary relationships, among which the mulberry DOF proteins were categorized into five groups (Group I, Group II, Group III, Group IV, and Group V). Bootstrap values are labeled above the branches. Different colors in the figure indicate different subgroups, and group name are marked on the periphery. Species abbreviations used: *Mn*, mulberry; *At*, *Arabidopsis*; *Os*, rice; *Vv*, grape; *Pp*, peach

tree. OG0003858 includes *MnDof4*, *ATDof3.4*, *ATDof5.8*, *PpeDof25*, *PpeDof24*, *VviDofL23*, *VviDofL24*, all of them are in the GroupIV of the tree. OG0008181 with *MnDof3*, *ATDof1.2*, *ATDof3.5*, *OsDof-6*, *OsDof-20*, *VviDofL10*. OG0008183 with *MnDof2*, *ATDof1.7*, *ATDof3.1*, *OsDof-21*, *PpeDof1*, *VviDofL2*, *VviDofL12*. OG0008548 with *MnDof1*, *ATDof1.5*, *ATDof2.3*, *VviDofL25*. OG0009925 with *MnDof10*, *ATDof2.1*, *ATDof3.2*, *ATDof5.3*, *VviDofL3*. OG0010537 with *MnDof20*, *OsDof-3*, *OsDof-24*, *VviDofL9*. OG0011914 with *MnDof13*, *ATDof5.6*, *PpeDof11*, *VviDofL5*. OG0012413 with *MnDof18*, *ATDof5.7*, *PpeDof8*, *VviDofL19*. OG0013101 with *MnDof11*, *ATDof5.4*, *PpeDof3*, *VviDofL12*. OG0013215 with *MnDof19*, *PpeDof9*, *VviDofL20*. OG0013691 with *MnDof12*, *PpeDof7*, *VviDofL21*. OG0013731 with *MnDof7*, *PpeDof10*, *VviDofL17*. OG0014309 with

MnDof8, *PpeDof2*. OG0014723 with *MnDof15*, *PpeDof14*. OG0015322 with *MnDof22*, *PpeDof23*. OG0017840 with *MnDof6*, *PpeDof16*.

Prediction of protein interactions

Protein–protein interaction (PPI) can help elucidate the mechanisms of action and regulatory mechanisms of the DOF protein family. Analysis of protein interaction patterns of mulberry DOF family genes (Fig. 5) using the STRING website predicted that 9 out of 24 DOF proteins interact with other mulberry proteins, among which MnDOF1 (XP_010103030.1) was predicted to interact with the growth hormone inducible protein aux22 (XP_010110304.1), linking Dof regulation to auxin-mediated cell elongation. MnDOF10 (XP_010104144.1) was predicted to interact with

Table 2 The orthogroups of *Dof* genes

Orthogroup ^a	Mulberry ^a	Arabidopsis ^a	Rice ^a	Peach ^a	Grape ^a
OG0000774 ^a	<i>MnDof16</i> , <i>MnDof17</i> , <i>MnDof14</i> ^a	<i>ATDOF2.4</i> , <i>ATDOF3.6</i> ^a	<i>OsDof-16</i> , <i>OsDof-19</i> , <i>OsDof-29</i> ^a	<i>PpeDof5</i> , <i>PpeDof6</i> , <i>PpeDof4</i> ^a	<i>VviDofL16</i> , <i>VviDofL14</i> , <i>VviDofL15</i> ^a
OG0001111 ^a	<i>MnDof9</i> , <i>MnDof5</i> ^a	<i>AtDOF1.8</i> , <i>AtDOF2.5</i> , <i>ATDOF3.7</i> , <i>ATDOF4.6</i> ^a	^a	<i>PpeDof15</i> , <i>PpeDof13</i> ^a	<i>VviDofL11</i> , <i>VviDofL13</i> , <i>VviDofL18</i> ^a
OG0001265 ^a	<i>MnDof21</i> , <i>MnDof24</i> , <i>MnDof23</i> ^a	<i>ATDOF3.3</i> <i>ATDOF5.2</i> ^a	<i>OsDof-11</i> ^a	<i>PpeDof22</i> , <i>PpeDof21</i> , <i>PpeDof20</i> ^a	<i>VviDofL7</i> , <i>VviDofL8</i> , <i>VviDofL6</i> ^a
OG0003858 ^a	<i>MnDof4</i> ^a	<i>ATDOF3.4</i> , <i>ATDOF5.8</i> ^a	^a	<i>PpeDof25</i> , <i>PpeDof24</i> ^a	<i>VviDofL23</i> , <i>VviDofL24</i> ^a
OG0008181 ^a	<i>MnDof3</i> ^a	<i>ATDOF1.2</i> , <i>ATDOF3.5</i> ^a	<i>OsDof-6</i> , <i>OsDof-20</i> ^a	^a	<i>VviDofL10</i> ^a
OG0008183 ^a	<i>MnDof2</i> ^a	<i>ATDOF1.7</i> , <i>ATDOF3.1</i> ^a	<i>OsDof-21</i> ^a	<i>PpeDof1</i> ^a	<i>VviDofL2</i> , <i>VviDofL12</i> ^a
OG0008548 ^a	<i>MnDof1</i> ^a	<i>ATDOF1.5</i> , <i>ATDOF2.3</i> ^a	^a	^a	<i>VviDofL25</i> ^a
OG0009925 ^a	<i>MnDof10</i> ^a	<i>ATDOF2.1</i> , <i>ATDOF3.2</i> , <i>ATDOF5.3</i> ^a	^a	^a	<i>VviDofL3</i> ^a
OG0010537 ^a	<i>MnDof20</i> ^a	^a	<i>OsDof-3</i> , <i>OsDof-24</i> ^a	^a	<i>VviDofL9</i> ^a
OG0011914 ^a	<i>MnDof13</i> ^a	<i>ATDOF5.6</i> ^a	^a	<i>PpeDof11</i> ^a	<i>VviDofL5</i> ^a
OG0012413 ^a	<i>MnDof18</i> ^a	<i>ATDOF5.7</i> ^a	^a	<i>PpeDof8</i> ^a	<i>VviDofL19</i> ^a
OG0013101 ^a	<i>MnDof11</i> ^a	<i>ATDOF5.4</i> ^a	^a	<i>PpeDof3</i> ^a	<i>VviDofL12</i> ^a
OG0013215 ^a	<i>MnDof19</i> ^a	^a	^a	<i>PpeDof9</i> ^a	<i>VviDofL20</i> ^a
OG0013691 ^a	<i>MnDof12</i> ^a	^a	^a	<i>PpeDof7</i> ^a	<i>VviDofL21</i> ^a
OG0013731 ^a	<i>MnDof7</i> ^a	^a	^a	<i>PpeDof10</i> ^a	<i>VviDofL17</i> ^a
OG0014309 ^a	<i>MnDof8</i> ^a	^a	^a	<i>PpeDof2</i> ^a	^a
OG0014723 ^a	<i>MnDof15</i> ^a	^a	^a	<i>PpeDof14</i> ^a	^a
OG0015322 ^a	<i>MnDof22</i> ^a	^a	^a	<i>PpeDof23</i> ^a	^a
OG0017840 ^a	<i>MnDof6</i> ^a	^a	^a	<i>PpeDof16</i> ^a	^a

growth hormone response factors (auxin response) factors (ARFs) (XP_010095167.1) and putative proteins (XP_010091212.1, XP_010092058.1, XP_010107919.1), MnDOF13 (XP_010101217.1) was also predicted to interact with ARFs (XP_010095167.1), as well as with showing co-expression with putative proteins (XP_010091212.1, XP_010092058.1, XP_010107919.1) MnDof10/13 co-target ARF5 (XP_010095167.1) with putative transcriptional repressors (XP_010091212.1), suggesting dual roles in auxin response fine-tuning. MnDOF16 (XP_010108748.1) and MnDOF17 (XP_010111483.1) were predicted to have a reciprocal relationship with Adagio protein 3 (XP_010088186.1). MnDOF21 (XP_010089911.1) was predicted to have reciprocal relationship with putative protein XP_010110817.1 and putative protein phosphatase 2C 35 (XP_010105687.1) and to interact with DNA binding protein SMUBP-2 (XP_010099518.1), and to be co-expressed with L-aspartate oxidase (XP_010108773.1) and zinc finger-like protein 9 (XP_010107355.1). This triad implies MnDOF21 may centrality in drought-stress cross-talk. MnDOF22 was predicted to interact with small ubiquitin-related modifier 2 (XP_010094465.1) and putative protein XP_010097648.1. MnDOF23 (XP_010108679.1) was predicted to interact with slow platelet protein 1 (XP_010094209.1), LHY protein (XP_010107134.1), two-component response regulator protein (XP_010099422.1), and slow plate protein 3 (XP_010088186.1). MnDOF24 (XP_010088142.1) was predicted to interact with small ubiquitin-related modifier 2 (XP_010094465.1) and putative protein XP_010097648.1, and to be co-expressed with DNA-binding protein SMUBP-2 (XP_010099518.1). MnDOF24 retains interactions with XP_010094465.1

and XP_010099518.1 across OG0001265 orthologs (Table 2), indicating conserved post-translational regulation mechanisms. These results providing a basis for further clarifying the mechanisms of action of mulberry DOF gene family members.

Chromosome localization analysis

Based on information on mulberry (*M. notabilis* L.) chromosomes [33], 24 *MnDof* genes were localized to specific mulberry chromosomes (Fig. 6). *MnDof* genes were distributed across all chromosomes, except for chromosomes 1, 2, and 7. *MnDof22* and *MnDof6* were located on chromosome 3; *MnDof1*, *MnDof2*, *MnDof7*, *MnDof8*, and *MnDof12* were located on chromosome 4, *MnDof3* and *MnDof4* were located on chromosome 5; *MnDof10* and *MnDof14* were located on chromosome 6; *MnDof11*, *MnDof13* and *MnDof21* were located on chromosome 8; *MnDof5* was located on chromosome 9; *MnDof24* was located on chromosome 10; *MnDof15*, *MnDof16* and *MnDof20* were located on chromosome 11; *MnDof9* and *MnDof17* were located on chromosome 12; *MnDof18* and *MnDof19* were located on chromosome 13; and *MnDof23* was located on chromosome 14.

Gene duplication events

To investigate evolutionary constraints and duplication events in the Dof gene family across species, collinearity analysis was performed to identify homologous gene pairs and infer selection pressures. We detected seven segmental duplication events among Dof family members (Table 3). All duplication events occurred on internal branches of the species tree (Non-Terminal type), indicating these duplications were shared by multiple

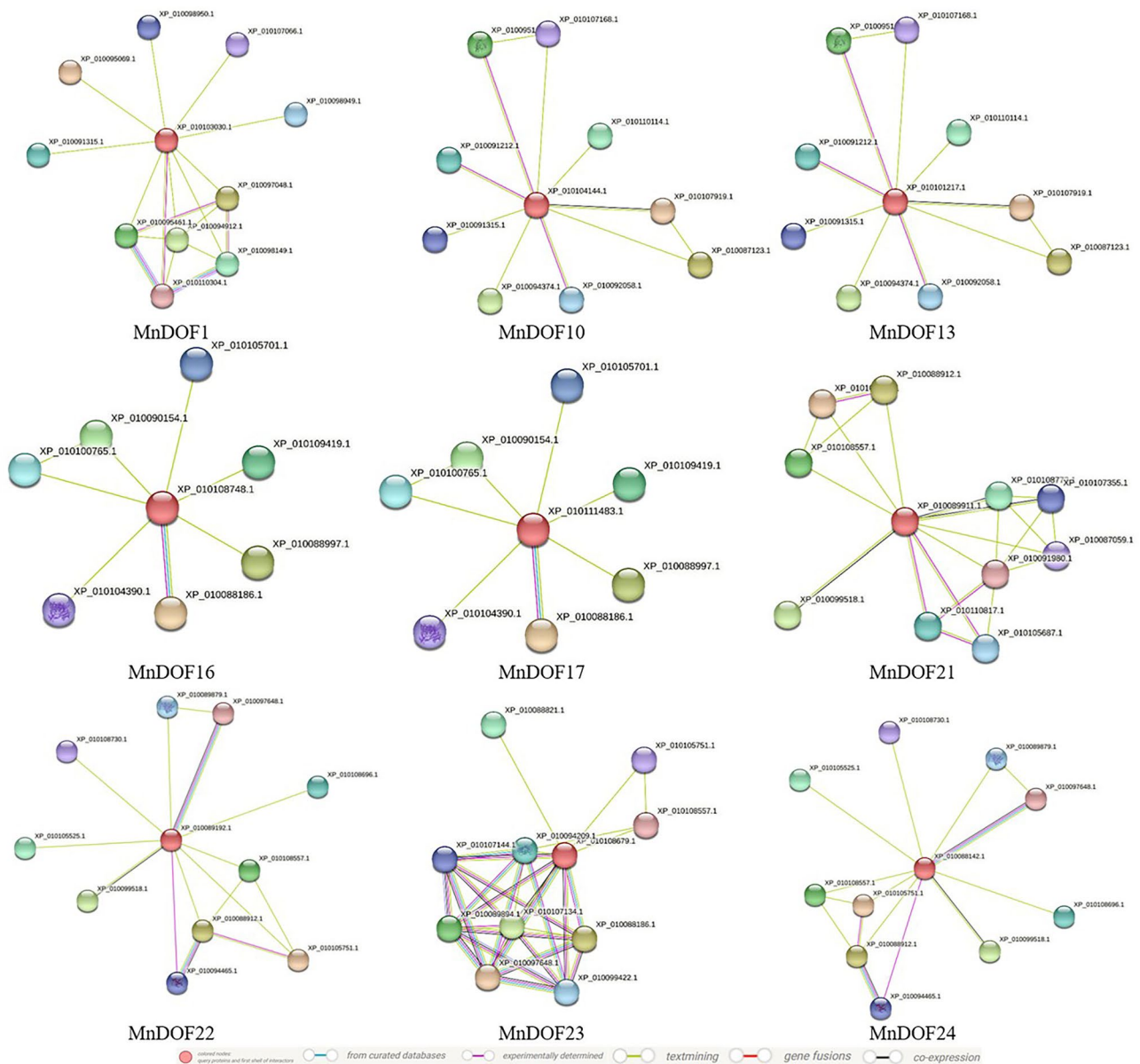


Fig. 5 Mulberry DOF transcription factor protein interactions

descendant lineages. Notably, orthogroup OG0001265 exhibited duplication events at two evolutionary nodes (N1 and N2), suggesting recurrent expansion during speciation.

The proteins sequence of Dofs was aligned by Mafft (v. 7.520). Then we converted the multiple sequence alignment of proteins and the corresponding mRNA sequences into a codon alignment by PAL2NAL (v. 14.1). The KaKs_Calculator (v. 2.0) was used for Ka and Ks estimation. The result was showed in Fig. 7.

Mulberry Dof gene expression under drought stress

In order to investigate the role of *MnDof* genes in mulberry drought stress, we detected changes in the expression

pattern of *MnDof* genes after PEG6000 treatment using qRT-PCR (Fig. 8). *MnDof2* and *MnDof22* were up-regulated by drought stress, *MnDof1*, *MnDof9*, *MnDof13*, *MnDof16*, and *MnDof24* were significantly down-regulated by drought induction ($P < 0.05$), *MnDof11* was up-regulated and then down-regulated, and *MnDof14* was not significantly changed. The above results indicated that the *MnDof* gene family members synergistically regulated the drought stress response of mulberry through differential expression patterns. Combined with the results of protein interactions (Fig. 5), we found that the up-regulated genes *MnDof2* and *MnDof22* were enriched in the stress signaling pathway, and *MnDof1* and *MnDof24* might be involved in the regulation of basal metabolism.

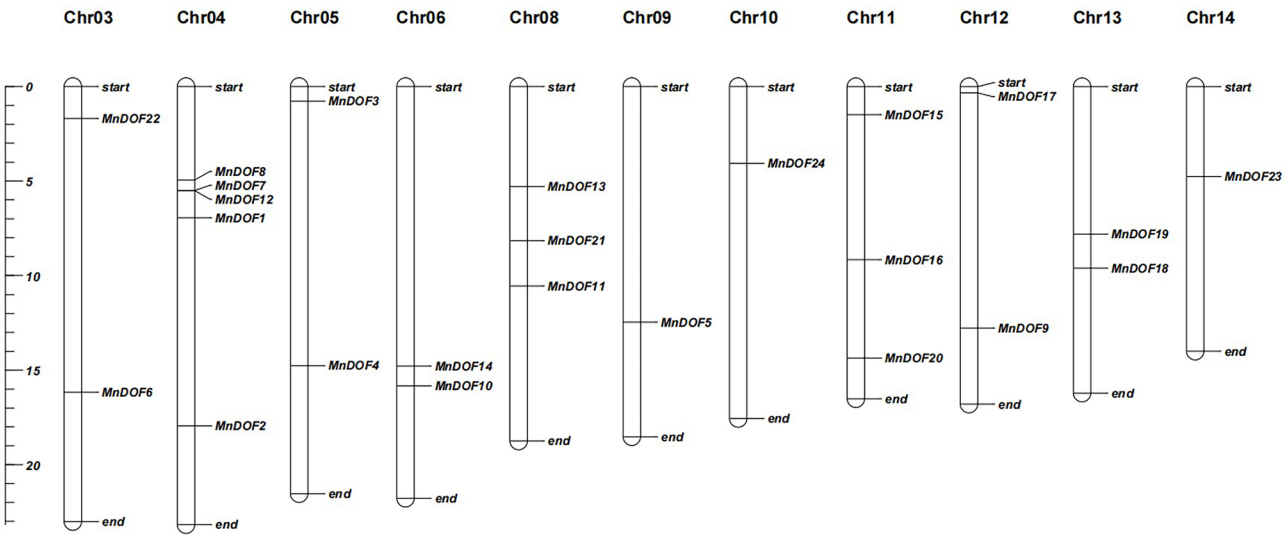


Fig. 6 Chromosomal localization of mulberry *Dof* genes

Table 3 The gene duplication events

Orthogroup ^a	Species Tree node ^a	Gene tree node ^a	Support ^a	Type ^a
OG0000774 ^a	N1 ^a	n3 ^a	1 ^a	Non-Terminal ^a
OG0000774 ^a	N1 ^a	n4 ^a	0.9 ^a	Non-Terminal ^a
OG0001111 ^a	N1 ^a	n0 ^a	1 ^a	Non-Terminal ^a
OG0001265 ^a	N1 ^a	n1 ^a	1 ^a	Non-Terminal ^a
OG0001265 ^a	N2 ^a	n12 ^a	1 ^a	Non-Terminal ^a
OG0003858 ^a	N2 ^a	n2 ^a	0.88889 ^a	Non-Terminal ^a
OG0008183 ^a	N0 ^a	n0 ^a	0.18182 ^a	Non-Terminal ^a

Note: Terminal: duplication on a terminal branch of the species tree; Non-Terminal: duplication on an internal branch of the species tree & therefore shared by more than one species^a

Discussion

Dof, a class of TFs that play important roles in plant growth and development, signal transcription, and stress tolerance, is present in many plants [34]. In this study, we systematically predicted and identified 24 non-redundant *Dof* genes in the mulberry genome (Table 1), which is comparable to the number of *Dof* genes identified in castor (24) [35], peach (25) [31], yellow vine (25) [36], grape (25) [15], sugarcane (25) [37], barley (26) [38], rice (30) [39], tomato (34) [40], cucumber (36) [41], and Arabidopsis (36) [42].

The present study showed that 24 mulberry *Dof* proteins differed in terms of coding region length, amino acid number, and theoretical isoelectric point, indicating that each protein likely performs different biological functions. Subcellular localization predictions showed that all mulberry *DOF* proteins localized to the nucleus, suggesting that they transcriptionally regulate downstream genes in the nucleus and play an important role in plant growth and development. Notably,

MnDof24 contains extended C-terminal motifs (Fig. 3) homologous to Arabidopsis MED15, suggesting co-activator functions in transcriptional complexes. A phylogenetic analysis showed that the mulberry *DOF* genes clustered into five groups with *DOF* genes in Arabidopsis, rice, grape, and peach, each represented by a specific single C2-C2 zinc finger structure, indicating that the members of each group have similar basic functions. The remaining nine motifs may be involved in other protein interactions that contribute to other life activities, such as regulation of cell development and growth in mulberry trees. However specific functions remain to be investigated in future studies.

DOF is a class of plant-specific TFs with important roles in growth and development, light regulation, hormone response, and stress response [43]. *AtDof5.8* in *Arabidopsis thaliana* promotes resistance to under drought and salt stress conditions [44]; *OsDof12* in rice can regulate the flowering period of rice under

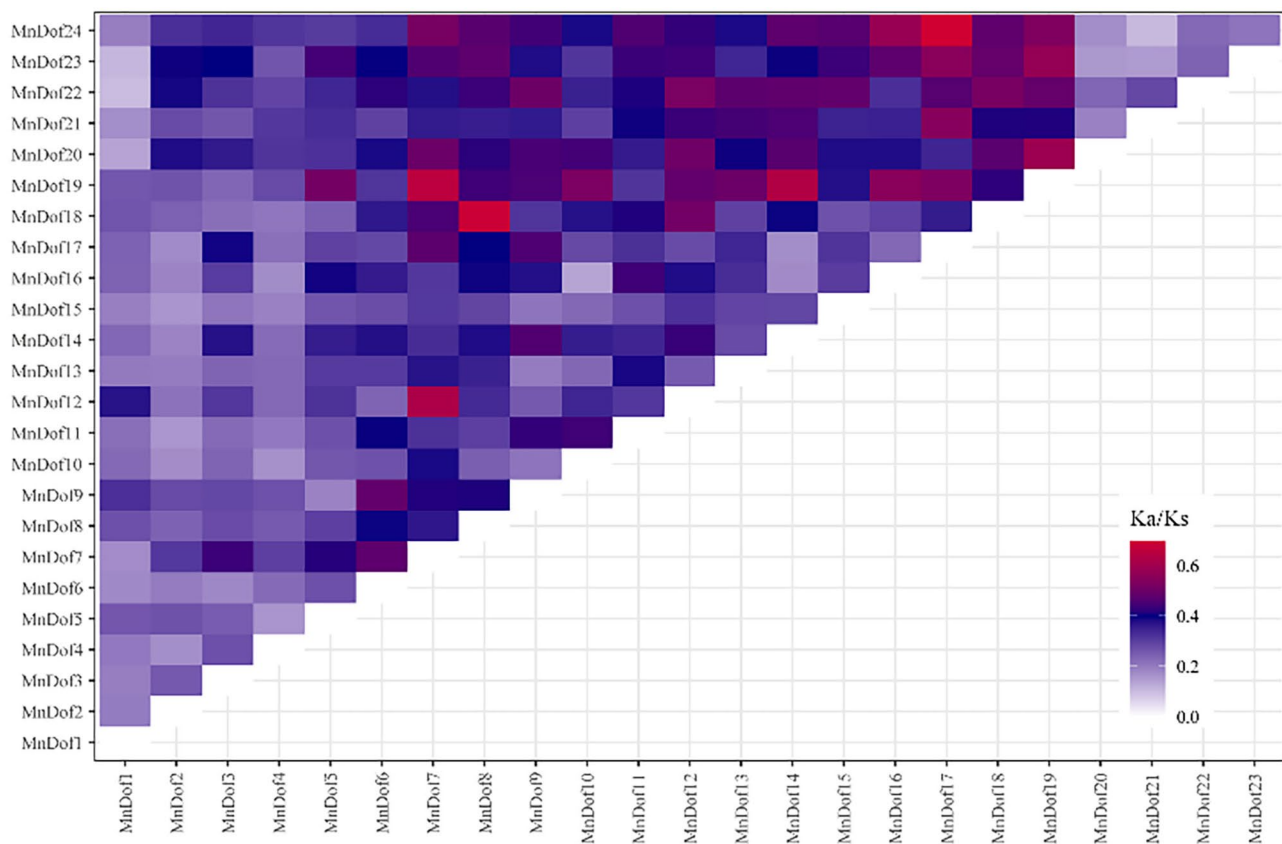


Fig. 7 The heatmap of pairwise ka/ks between Dof genes

long sunlight conditions [45]; *ZmDof1* in maize can suppress the expression of *Zm401* to regulate pollen development before flowering in maize [46]; and the *VvDof3* gene in grapes can be induced by jasmonic acid and salicylic acid induced and enhanced resistance to powdery mildew through the salicylic acid signaling pathway [47]; *AtDof2.1* (*AT2G28510.1*) is an upstream regulator of the jasmonate hormone regulatory pathway and can directly bind to the promoter of *MYC2* to affect downstream gene expression [48]. Predictive analysis of cis-acting promoter elements is an important tool for studying gene function, and, in this study, we found that mulberry *Dof* family genes contain many elements related to adversity stress and hormone responses, implying that *Dof* genes play important roles in resisting abiotic stress and responding to hormone metabolism. *MnDof22* contains three drought-responsive elements (DRE/MBS, etc.), which are associated with its up-regulation under PEG6000 treatment, and it could be used as a molecular marker to assist in screening for drought-resistant germplasm resources in future studies. Protein interaction analyses showed that some *Dof* genes in mulberry are co-expressed with various proteins and play important roles in life activities such as protein composition and

folding. *MnDof23* interacts with the biological clock protein LHY (Fig. 5), coupling photoperiodic and drought-responsive pathways, which can be designed to be a light-regulated drought-resistant gene circuit in future study.

Many studies have shown that adversity stress alters the expression pattern of plant *Dof* genes, which are essential for plant stress tolerance. In this study, nine *Dof* genes responded to drought stress, suggesting that the *MnDof* genes may play a role in promoting drought tolerance in mulberry under harsh environments. In *Arabidopsis thaliana* and tomato, cation diffusion facilitators (CDFs) not only regulate photoperiod and flowering, but are also participate in the response to desiccating salt, drought, high temperature, and abscisic acid induction, and plants that overexpressing these factors appear more resistant to drought and flower later [27, 47, 49]. *MnDof1*, homologous to the CDF gene of *Arabidopsis thaliana*, is responsive to drought stress, and *MnDof1*, *MnDof23*, and *MnDof24* belong to the same subfamily (group V). The results of this study lay the foundation for further in-depth studies on the function of mulberry *Dof* gene family members and provide ideas for genetic screening for

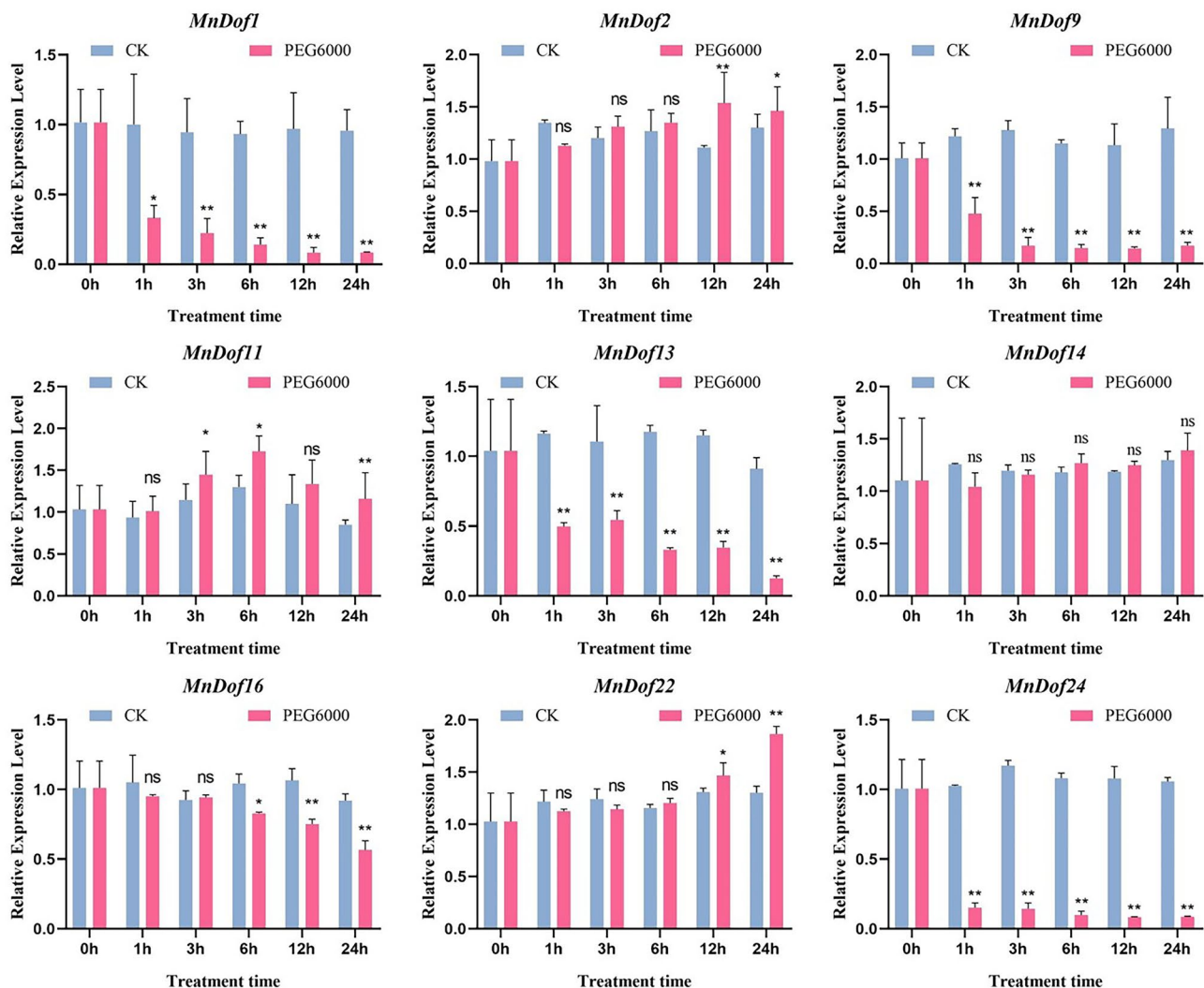


Fig. 8 Expression of 9 Dof genes in mulberry tree under drought stress

selection and breeding of varieties in mulberry grown in extreme environments.

Conclusions

In this study, 24 non-redundant *MnDof* gene family members were identified in the mulberry genome, and all of the encoded proteins were predicted to contain highly conserved C2-C2 type *Dof* zinc finger structures. *Dof* gene family members were divided into five subgroups according to their evolutionary relationships and structural features, and the *MnDof* genes in each subgroup showed similar gene structures and conserved protein domains. Responses of some mulberry *Dof* genes to drought stress were apparent, suggesting that members of the *MnDof* gene family synergistically regulate the growth and development of mulberry trees and participate in their responses to adversity stress, especially drought stress.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12864-025-11843-z>.

Supplementary Material 1.
Supplementary Material 2.
Supplementary Material 3.

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Authors' contributions

JD: Performed the experiments; Analyzed and interpreted the data, wrote the paper. Y.L.W: Analyzed and interpreted the data, revision of papers. J.L, L.W, Y.C.Z: Performed the experiments; Analyzed and interpreted the data. S.M.L: Analyzed and interpreted the data. J.C.Z: performed the statistical analysis. G.Q.H and G.L: Conceived and designed the experiments; revised the manuscript. All authors read and approved the final manuscript.

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Data availability

Data is provided within the manuscript or supplementary information files. The full-length genome sequence of *M. notabilis* and its protein sequences were obtained from the NCBI database (https://www.ncbi.nlm.nih.gov/data-hub/genome/GCF_000414095.1/) and the Genome Database published by Southwest University (<https://morus.biodb.org/morusdb/>). The assembly of *Morus alba* (GWHDOOO000000000), *Morus alba* var. *atropurpurea* (GWHBOTE000000000), *Morus yunnanensis* (GWHBISP000000000), and The T2T assembly of *Morus mongolica* (GCA_045269455.1) and *Morus notabilis* (GWHCBHX000000000) were downloaded according to the accession number. The protein sequence of *Arabidopsis* (*Arabidopsis thaliana*), rice (*Oryza sativa Japonica*), grape (*Vitis vinifera*), and peach (*Prunus persica*) were download from EnsemblPlants (<https://plants.ensembl.org/index.html>). Public data of mulberry drought-related transcriptomes (SRR12513708, SRR12513709, SRR12513710, SRR12513711).

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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